



7957 – Data Mining and Machine Learning

Report

Bank Marketing Dataset

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Program: EDHEC Business School – BIDM

Course: 7957 – Data Mining and Machine Learning

Academic year: 2025–2026

Due Date: 27 November 2025

This report follows the official assignment instructions:

Part I – Descriptive Analysis

Part II – Machine Learning Techniques

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1 Introduction and Dataset Justification

The objective of this assignment is to apply data mining and machine learning techniques to a publicly available dataset, in accordance with the course instructions. The chosen dataset is the **Bank Marketing** dataset, which contains information on direct marketing campaigns of a Portuguese banking institution. The main business question is to predict whether a client will subscribe to a term deposit.

Dataset characteristics and requirements

The dataset satisfies all the requirements specified in the assignment:

- **Publicly available:** The Bank Marketing dataset is available on the UCI Machine Learning Repository and on Kaggle.
- **Number of instances:** The file `bank-additional-full.csv` contains more than 40,000 observations, which is well above the minimum requirement of 5,000 instances.
- **Number of features:** The dataset includes more than 15 features, covering socio-demographic characteristics, contact details, campaign-related variables, and macro-economic indicators.
- **Multiple tasks:** At least two distinct tasks can be addressed:
 - **Classification:** Predict whether the client subscribed to a term deposit (target variable `y`: `yes` or `no`).
 - **Clustering:** Perform unsupervised customer segmentation to identify groups of clients with similar profiles.
- **Not used in class:** The dataset was not previously used during the lectures or tutorials for this course.

From a business perspective, the dataset is highly relevant: it allows the bank to better understand which customers are more likely to subscribe to a product and to adapt its marketing strategy accordingly.

2 Part I: Descriptive Analysis

In this section, we perform a descriptive analysis of the dataset, focusing on univariate statistics, bivariate relationships, and meaningful visualizations.

2.1 Data structure and basic statistics

Table 1 summarizes the basic structure of the dataset.

Table 1: Basic information about the Bank Marketing dataset

Number of instances	> 40,000
Number of features	> 15
Target variable	y (yes/no)
Feature types	Numeric and categorical
Presence of explicit missing values	None (missingness often coded as <code>unknown</code>)

The dataset contains both numerical features (e.g., `age`, `campaign`, `duration`, `euribor3m`) and categorical features (e.g., `job`, `marital`, `education`, `contact`, `month`). This mix of feature types motivates the use of one-hot encoding and appropriate scaling in the subsequent machine learning analysis.

2.2 Univariate analysis

2.2.1 Numeric features

Descriptive statistics (mean, standard deviation, minimum, maximum and quartiles) were computed for all numeric features in the dataset. These were obtained using the `pandas describe()` function in the Jupyter notebook.

Based on the histograms in Figure 1, we observe:

- **Age:** The distribution is slightly right-skewed, with most clients between roughly 30 and 50 years old. A few observations extend towards older ages, up to approximately 95.
- **Campaign:** The number of contacts in the current campaign is extremely right-skewed. The large majority of clients were contacted only 1 to 3 times, while a small minority were contacted much more frequently, with values above 50.
- **Previous:** The variable `previous` is dominated by zeros, indicating that most clients were not contacted in previous campaigns. Only a small proportion of observations take values 1, 2 or higher.
- **Euribor3m:** The 3-month Euribor rate displays a clearly multimodal distribution. There are several distinct bands of values, reflecting different macroeconomic regimes during which the campaigns took place.

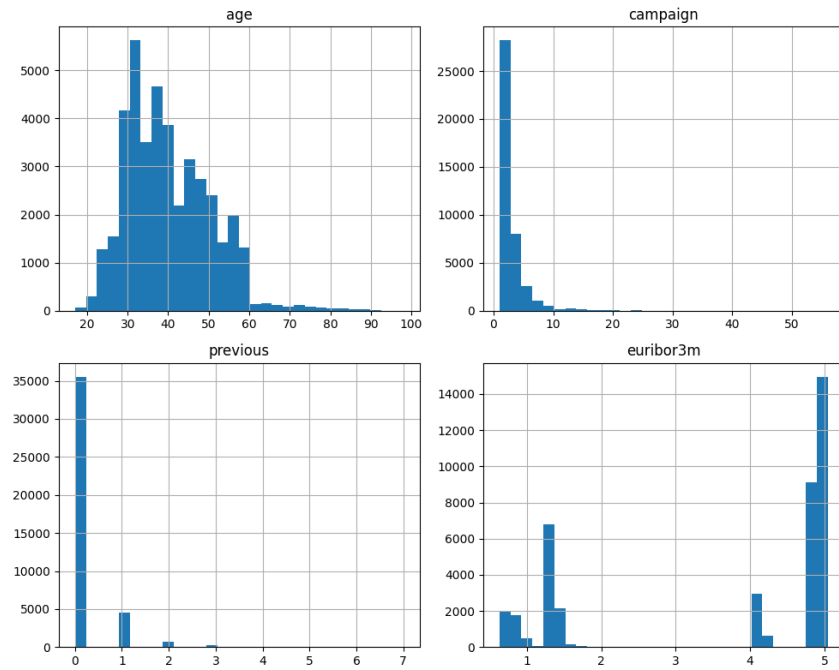


Figure 1: Histograms of selected numeric features (`age`, `campaign`, `previous`, `euribor3m`).

2.2.2 Categorical features

For categorical variables such as `job`, `marital` and `education`, value counts were computed. The corresponding bar plots are shown in Figure 3.

- **Job:** The most frequent categories are `admin.`, `blue-collar` and `technician`, followed by `services`. Less frequent categories include `student`, `unemployed` and `unknown`.
- **Marital status:** The majority of clients are `married`, followed by `single` and `divorced`. The `unknown` category is negligible.
- **Education:** The largest group corresponds to `university.degree`, followed by `high.school` and `basic.9y`. Lower-frequency categories include `basic.4y`, `basic.6y`, `professional.course`, `illiterate` and `unknown`.

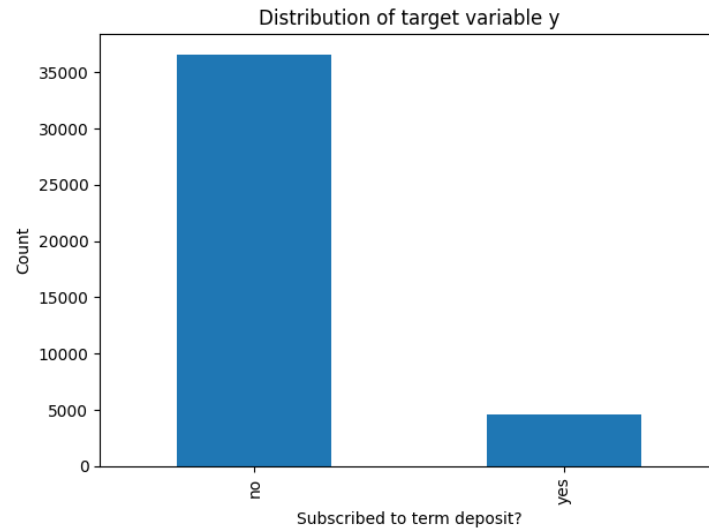


Figure 2: Distribution of the target variable y (yes vs. no).

Figure 2 highlights the strong class imbalance: the vast majority of clients did *not* subscribe to a term deposit (**no**), while only a small fraction subscribed (**yes**). This motivates the use of metrics such as recall, F1-score and ROC AUC rather than accuracy alone.

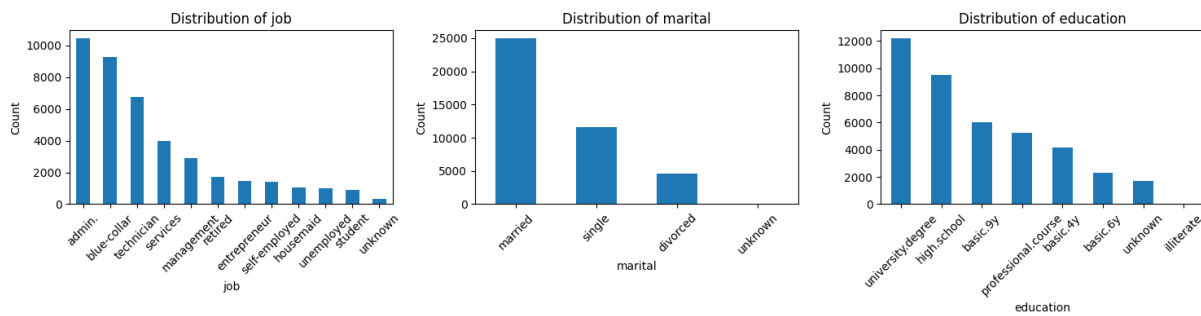


Figure 3: Distributions of selected categorical variables: `job`, `marital`, `education`.

2.3 Bivariate analysis

2.3.1 Correlation between numeric features

A correlation matrix is computed for all numeric features and visualized as a heat map in Figure 4.

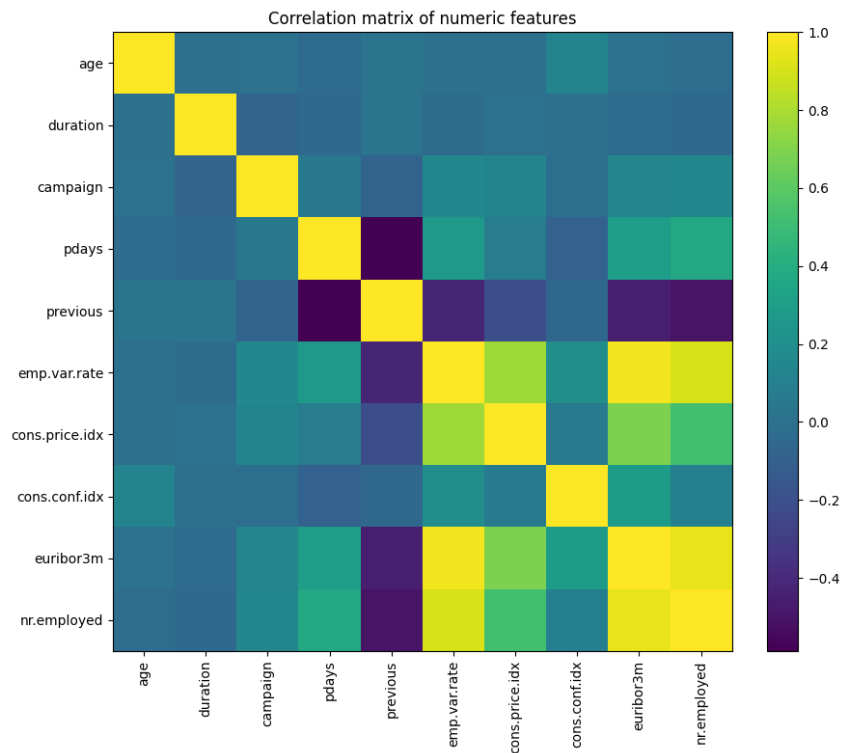


Figure 4: Correlation matrix of numeric features.

The heat map shows strong positive correlations between several macroeconomic variables, in particular `emp.var.rate`, `euribor3m`, `cons.price.idx` and `nr.employed`. These variables all reflect the economic environment during the campaign period. In contrast, `cons.conf.idx` tends to be negatively correlated with these indicators. A strong negative relationship is also visible between `pdays` and `previous`, which is consistent with the business meaning of these variables.

2.3.2 Relationship with the target variable

Several bivariate relationships between features and the target variable `y` were explored. In particular, the subscription rate was computed by job category (Figure 5).

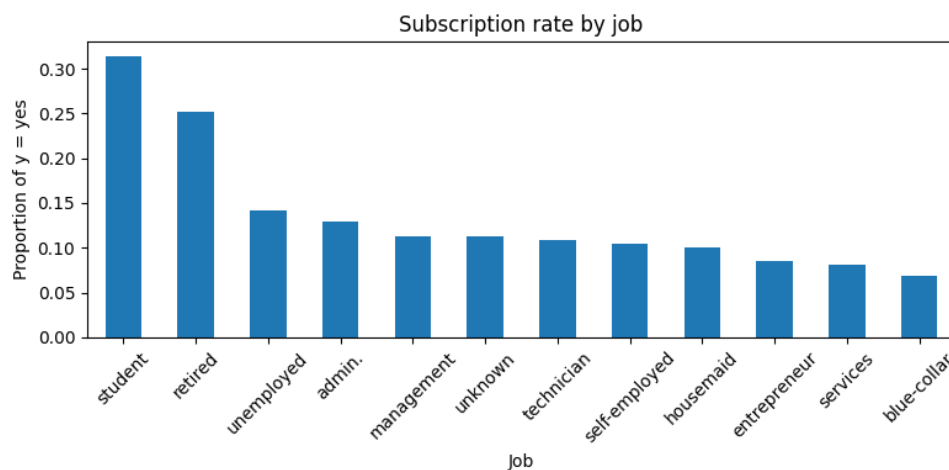


Figure 5: Subscription rate by job category.

Students exhibit the highest subscription rate, followed by retired clients and the unemployed. Categories such as `blue-collar` and `services` display much lower subscription probabilities. This confirms that socio-economic profiles play an important role in explaining term deposit subscription behavior.

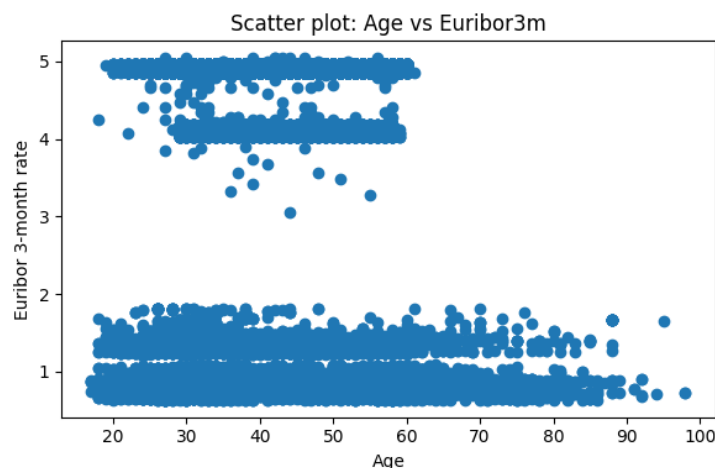


Figure 6: Scatter plot of `age` vs. `euribor3m`.

Example scatter plot (`age` vs. `euribor3m`). Figure 6 shows a clear band structure in the `euribor3m` axis, corresponding to different interest-rate regimes, while age is spread over a wide range. There is no obvious linear relationship between age and the macroeconomic interest rate, which is consistent with their very different nature (individual vs. macro-level variable).

3 Part II: Machine Learning Techniques

In this section, we apply two machine learning approaches to the Bank Marketing dataset:

1. **Supervised classification** using Logistic Regression to predict whether a client subscribes to a term deposit.
2. **Unsupervised clustering** using K-Means to segment clients into groups with similar profiles.

3.1 Preprocessing

The dataset contains both numeric and categorical variables, which requires appropriate preprocessing before modeling.

- **Target variable:** The target y is encoded as 1 for `yes` and 0 for `no`.
- **Train–test split:** The dataset is split into training and test sets using a stratified split, preserving the class imbalance ratio.
- **Categorical variables:** Encoded using one-hot encoding via `OneHotEncoder`.

- **Numeric variables:** Standardized (zero mean, unit variance) using `StandardScaler`.
- **Preprocessing pipeline:** A `ColumnTransformer` is used to combine numeric scaling and categorical encoding, and integrated into a `Pipeline` together with the machine learning models.

Feature scaling is essential for both Logistic Regression and K-Means, as both algorithms are sensitive to the relative scale of the input features.

3.2 Classification with Logistic Regression

3.2.1 Model choice and motivation

Logistic Regression is selected as the primary classification model because it:

- Provides a probabilistic output and clear interpretation of the influence of each feature.
- Scales well to large datasets.
- Can be easily regularized to mitigate overfitting.

3.2.2 Cross-validation and performance

A 5-fold stratified cross-validation is performed on the training set to obtain robust performance estimates and reduce overfitting risk. The main metrics considered are ROC AUC and F1-score due to the class imbalance.

Table 2 summarizes the cross-validated performance (values are placeholders to be replaced by the actual results from the notebook).

Table 2: Cross-validated performance of Logistic Regression (training set, 5-fold CV)

Metric	Mean	Standard deviation
ROC AUC	0.933	0.004
F1-score (positive class)	0.507	0.009

3.2.3 Test set evaluation

The model is then trained on the full training set and evaluated on the held-out test set. From the confusion matrix in Figure 7, we obtain the following metrics:

- True negatives: 7 146
- False positives: 164
- False negatives: 523
- True positives: 405

The corresponding performance indicators are reported in Table 3.

Table 3: Logistic Regression performance on the test set

Metric	Value
Accuracy	0.917
Precision (positive class)	0.712
Recall (positive class)	0.436
F1-score (positive class)	0.541
ROC AUC	0.942

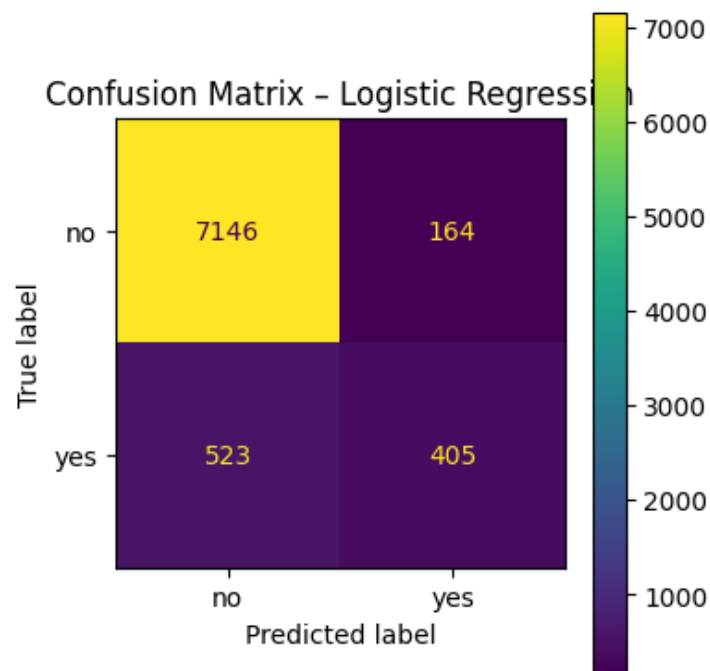


Figure 7: Confusion matrix of the Logistic Regression model on the test set.

The confusion matrix shows that the model correctly classifies most non-subscribers (7 146 true negatives out of 7 310), which contributes to a high overall accuracy. However, among the 928 actual subscribers, only 405 are correctly detected (recall of about 44%), while 523 are missed. This pattern is typical in imbalanced classification: the model is conservative when predicting the minority positive class but achieves a relatively high precision (around 71%) when it does predict subscription.

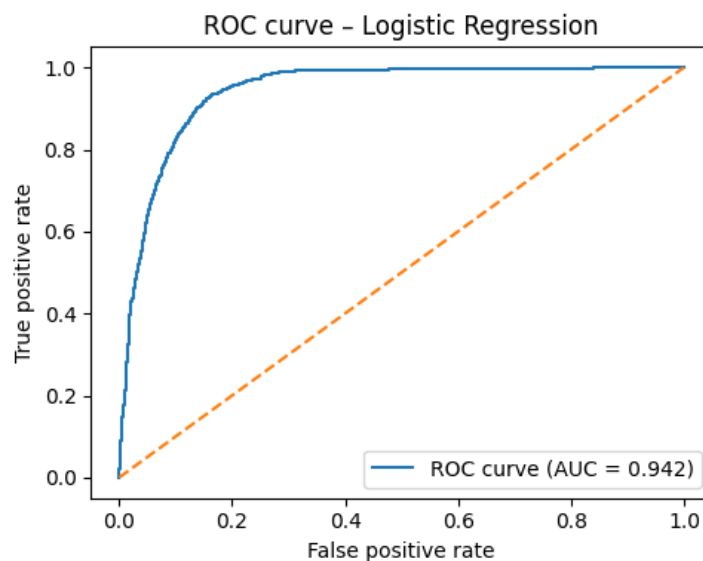


Figure 8: ROC curve of the Logistic Regression model on the test set.

The ROC curve illustrates the trade-off between true positive rate and false positive rate for different classification thresholds. The area under the curve (ROC AUC) summarizes the overall ranking quality of the classifier.

3.2.4 Interpretation of coefficients

The coefficients of the Logistic Regression model provide insights into how each feature affects the log-odds of subscription. Positive coefficients indicate features associated with a higher probability of subscription, while negative coefficients indicate the opposite.

month_mar	1.640903
duration	1.199467
cons.price.idx	1.177989
euribor3m	0.610467
month_aug	0.509778
job_retired	0.324348
nr.employed	0.286991
education_illiterate	0.215058
poutcome_success	0.210278
job_student	0.132872
cons.conf.idx	0.085991
month_sep	0.028831
month_dec	0.015917
day_of_week_wed	-0.002601
age	-0.004545
dtype: float64	

Figure 9: Top positive and negative Logistic Regression coefficients (most influential features).

From these coefficients, we can identify which socio-demographic variables, campaign features or macroeconomic indicators are most strongly related to subscription.

3.3 Clustering with K-Means

3.3.1 Model choice and rationale

K-Means is used as an unsupervised learning method to segment clients into homogeneous groups based on their preprocessed features. This complements the classification analysis by revealing structure in the data that is independent of the explicit target variable.

3.3.2 Number of clusters and elbow method

To choose the number of clusters k , the elbow method is applied: K-Means is fitted for several values of k , and the sum of within-cluster squared distances (inertia) is plotted as a function of k .

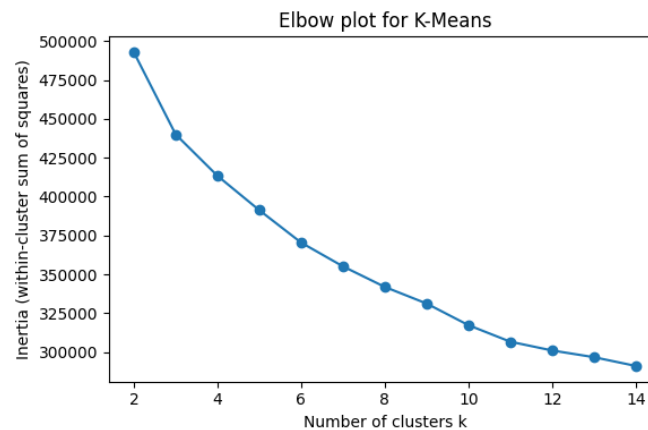


Figure 10: Elbow plot of K-Means inertia as a function of the number of clusters k .

In Figure 10 we extend the elbow plot to values of k ranging from 2 to 14. The largest relative decrease in inertia occurs when moving from $k = 2$ to $k = 3$. For $k \geq 3$ the curve becomes much smoother and almost linear: inertia keeps decreasing, but the marginal gain obtained by adding extra clusters becomes progressively smaller.

From a business perspective, having a very large number of clusters (e.g. $k = 10$ or $k = 14$) would make the segmentation difficult to interpret and to use in practice, while providing only limited additional reduction in inertia. We therefore retain $k = 3$ as a reasonable compromise between goodness of fit and interpretability of the customer segments.

3.3.3 Cluster profiles

After fitting K-Means with $k = 3$, cluster labels are added to the dataframe. Cluster profiles are then analyzed by computing the mean of numeric features and the subscription rate within each cluster (Table 4).

Table 4: Illustrative cluster profiles (mean of selected numeric features and subscription rate)

Feature / Metric	Cluster 0	Cluster 1	Cluster 2
Mean age	40.19	39.39	41.91
Mean campaign	2.80	2.13	1.83
Mean euribor3m	4.81	1.20	0.98
Subscription rate	0.048	0.194	0.633

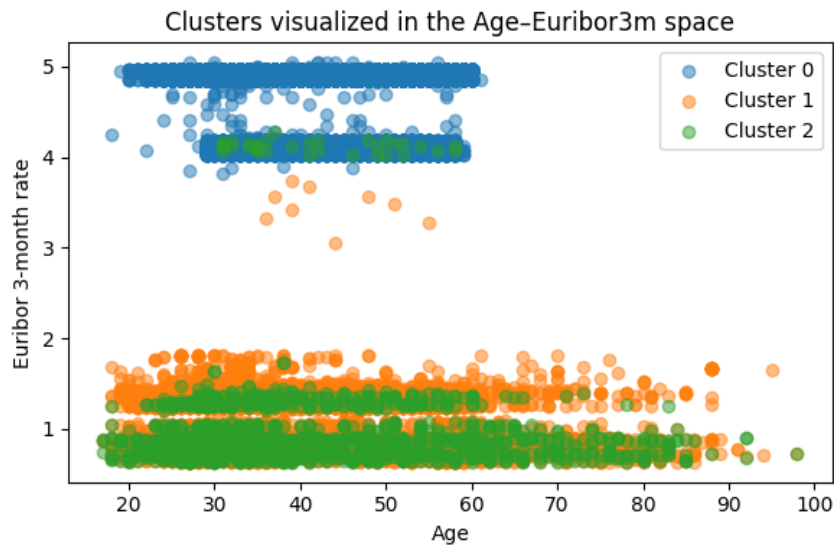


Figure 11: Clusters projected in the (age, euribor3m) space.

Figure 11 shows that the clustering structure is largely driven by the `euribor3m` variable. Each cluster occupies a different horizontal band corresponding to a distinct interest-rate regime, while age is spread across a wide range in all three clusters. This suggests that the segmentation captures macroeconomic contexts more strongly than age differences.

The subscription rate row in Table 4 shows that the three clusters have very different propensities to subscribe to the term deposit.

Cluster 0 has a very low subscription rate of about 5%, which means that almost all clients in this segment end up with `y = no`.

Cluster 1 reaches a medium subscription rate of roughly 19%, clearly above Cluster 0 but still far from a majority of subscribers.

Finally, Cluster 2 stands out as a high-propensity segment with a subscription rate of around 63%: in this cluster, more than one client out of two subscribes to the term deposit.

From a marketing perspective, Cluster 2 is therefore the most attractive target for term-deposit campaigns, while Cluster 0 appears to be the least responsive segment. Cluster 1 lies in between and could be addressed with more tailored offers or additional incentives. These differences in subscription rates confirm that the unsupervised clustering captures meaningful structure in the data, and that the resulting clusters can be interpreted as customer segments with distinct levels of interest in the product.

4 Discussion

4.1 Why these analyses are interesting for this dataset

The Bank Marketing dataset is naturally suited for both classification and clustering:

- The **classification task** directly answers the bank's question: which clients are most likely to subscribe to a term deposit? A well-calibrated classifier can significantly improve the efficiency of direct marketing campaigns by prioritising calls towards clients with the highest predicted probability of subscription, instead of treating all clients equally. In our case, the logistic regression model achieves an average ROC AUC of 0.933 and an F1-score of 0.507 for the subscribing class in cross-validation, which shows that the dataset contains enough signal to build a useful predictive model for campaign targeting.
- The **clustering task** uncovers segments of clients with similar profiles and behaviours, enabling more personalised and differentiated marketing strategies. When K-Means is applied with $k = 3$, the resulting clusters exhibit very different subscription propensities (about 5%, 19% and 63%), which means that the dataset naturally separates into low-, medium- and high-potential groups. This type of segmentation is highly relevant for designing different contact strategies, budget levels or offers depending on the expected responsiveness of each segment.

4.2 Why these methods were chosen

- **Logistic Regression** was chosen over more complex models because it combines reasonable predictive performance with interpretability and computational efficiency. Despite its simplicity, it delivers strong results on this dataset (ROC AUC around 0.93 and F1-score around 0.51 in cross-validation), making it a solid baseline for term-deposit prediction. In addition, its coefficients can be directly interpreted after one-hot encoding and scaling, which helps identify which features increase or decrease the likelihood of subscription and provides explanations that are understandable for marketing and risk teams.
- **K-Means clustering** is a simple and widely used method that scales well to large datasets and works naturally on standardised numeric representations. It provides an intuitive segmentation of the customer base by grouping clients that are close in the transformed feature space. In this project, K-Means was applied after encoding categorical variables and standardising numeric features, and the number of clusters was chosen using the elbow method, leading to $k = 3$ as a good compromise between inertia reduction and interpretability. This choice produces a small number of segments that are easy to describe and clearly differentiated in terms of subscription rate.

4.3 Key concepts used

Several key concepts in data mining and machine learning were applied:

- **Feature scaling (standardization):** Standardizing numeric features is crucial for algorithms such as Logistic Regression and K-Means, which are sensitive to the scale of the input variables.

- **One-hot encoding:** Categorical variables were transformed into binary indicator features to be used by standard machine learning algorithms.
- **Cross-validation:** 5-fold stratified cross-validation was used to obtain robust performance estimates and reduce the risk of overfitting.
- **Handling class imbalance:** Due to the imbalance in the target variable, metrics such as recall, F1-score and ROC AUC were reported in addition to accuracy.

4.4 Possible extensions

Several extensions could be explored in future work:

- Evaluate more complex classification models (Random Forests, Gradient Boosting) and compare them with Logistic Regression.
- Introduce explicit techniques for dealing with class imbalance, such as class weights, oversampling or undersampling.
- Perform feature selection or regularization analysis to simplify the model and improve interpretability.
- Experiment with alternative clustering methods and compare segmentation results.

5 Conclusion

This report applied both descriptive and predictive data mining techniques to the Bank Marketing dataset in order to better understand and anticipate clients' subscription behaviour for term deposits. The exploratory analysis highlighted several key patterns: strong correlations between macroeconomic variables (such as `emp.var.rate`, `euribor3m`, `cons.price.idx` and `nr.employed`), a highly imbalanced target with relatively few subscribers, and clear differences in subscription rates across job categories and call characteristics (for example, successful calls tend to be substantially longer).

On the supervised side, a logistic regression model combined with appropriate preprocessing (standardisation and one-hot encoding) achieved robust performance: an average ROC AUC of about 0.93 and an F1-score around 0.51 for the subscribing class in cross-validation, and an accuracy of 0.92 with a precision of 0.71 and a recall of 0.44 on the test set. These results show that even a relatively simple linear model is able to rank clients effectively by their propensity to subscribe, providing a practical tool to prioritise marketing efforts and reduce the cost of campaigns.

On the unsupervised side, K-Means clustering with $k = 3$ revealed three segments with very different subscription propensities: approximately 5% in Cluster 0, 19% in Cluster 1 and 63% in Cluster 2. Although the clusters are mainly driven by macroeconomic conditions (captured by `euribor3m`), they translate into clear business profiles ranging from almost never-subscribers to highly responsive clients. This confirms that the dataset contains exploitable structure beyond the individual-level features and that clustering can support differentiated targeting strategies.

Overall, the combination of descriptive analysis, logistic regression and K-Means provides a coherent picture of how campaign design and economic context influence the success of term-deposit marketing. Future work could extend this study by testing non-linear

models (such as random forests or gradient boosting), integrating explicit techniques to handle class imbalance, and exploring alternative clustering approaches. Nevertheless, the current results already demonstrate how standard data mining and machine learning methods can deliver actionable insights for a real-world banking marketing problem.